

NovaCart

India's Intelligent Online Marketplace

Smart Shopping. Smarter Decisions.

Business Questions Roadmap

Profit Recovery Analytics Platform

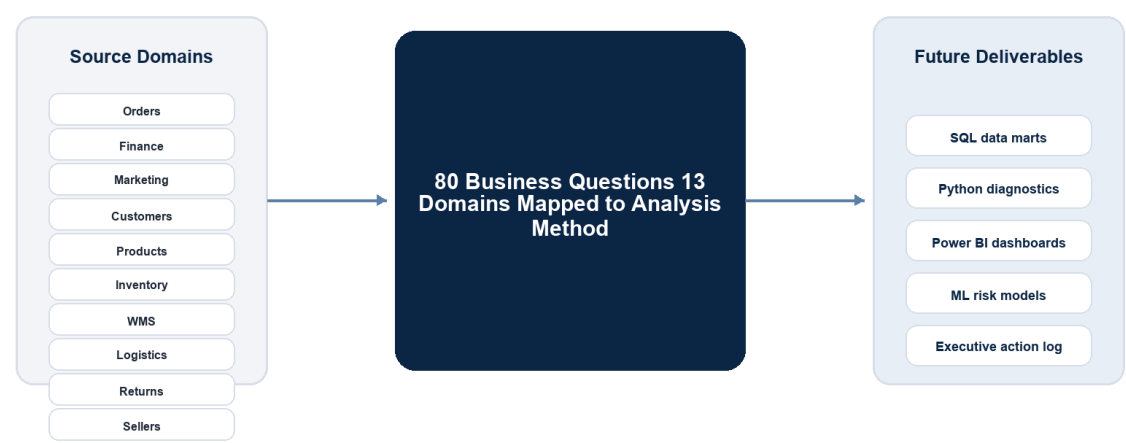
Roadmap Detail	Value
Deliverable	05_Business_Questions
Version	1.0
Period	FY2025-FY2026
Prepared by	Aishi Dutta
Question roadmap	80 questions across 13 business domains
Analysis methods	SQL, Python, Power BI, Machine Learning

Purpose. This document defines the analysis questions that will guide future SQL datasets, Python diagnostics, Power BI dashboards and Machine Learning exploration.

Roadmap Architecture

The roadmap converts NovaCart's approved business context, stakeholder needs and KPI dictionary into analysis-ready questions. Each question is assigned to the most appropriate delivery method so downstream work can be sequenced without redefining the business problem.

Business Questions Roadmap



Roadmap principle: each question translates an executive or functional decision need into a future SQL, Python, Power BI or Machine Learning work item.

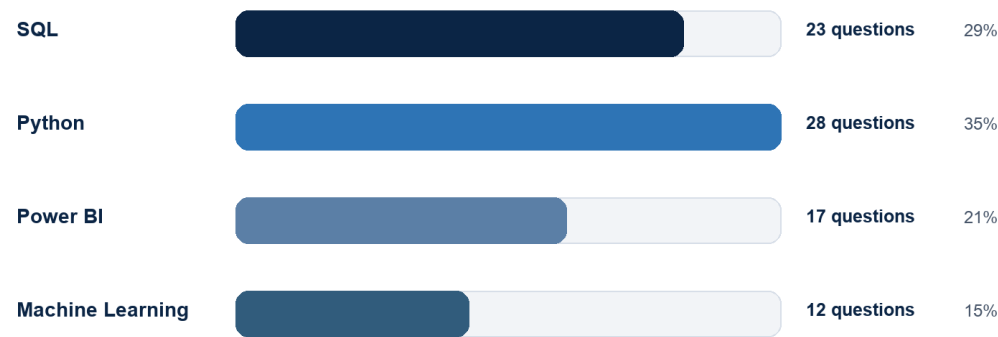
Figure 1. Business questions roadmap from source domains to future analysis deliverables.

Business Domain	Questions	Use
Sales	7	Roadmap question group
Revenue	6	Roadmap question group
Profitability	7	Roadmap question group
Marketing	6	Roadmap question group
Customer Behaviour	6	Roadmap question group
Product Performance	6	Roadmap question group
Inventory	6	Roadmap question group
Warehouse	6	Roadmap question group
Logistics	6	Roadmap question group
Returns	6	Roadmap question group
Finance	6	Roadmap question group
Seller Performance	6	Roadmap question group
Executive Reporting	6	Roadmap question group

Analysis Method Assignment

Method assignment is based on the question's intended analytical job. Extraction and reconciliation questions are assigned to SQL, diagnostic and scenario questions to Python, recurring consumption questions to Power BI, and prediction or ranking questions to Machine Learning.

Analysis Method Coverage



Method mix intentionally separates data extraction, diagnostic analysis, dashboard consumption and predictive modelling.

Figure 2. Coverage of roadmap questions by primary analysis method.

Method	Roadmap role
SQL	Data extraction, joins, segmentation, reconciliation and metric validation.
Python	Exploratory diagnostics, driver analysis, scenarios, cohort analysis and financial bridges.
Power BI	Recurring management views, scorecards, drilldowns and operating review dashboards.
Machine Learning	Prediction, prioritisation, risk scoring, ranking and early-warning models.

- The roadmap contains business questions only; implementation code belongs in later SQL, Python, ML and Power BI deliverables.
- Every downstream artifact should reference the same company facts, stakeholder needs and KPI definitions already established in prior NovaCart documents.

Sales and Revenue Questions

These questions define where demand and revenue quality are coming from.

ID	Domain	Business Question	Method	Roadmap Use
BQ-SAL-01	Sales	Which regions, customer segments and categories are driving the highest order growth?	SQL	Prioritize growth segments and sales focus.
BQ-SAL-02	Sales	Which product categories show rising order volume but weak average order value?	SQL	Separate volume growth from basket quality.
BQ-SAL-03	Sales	Which customer cohorts contribute the largest share of repeat orders?	Python	Target retention actions to profitable cohorts.
BQ-SAL-04	Sales	Which seller groups generate strong order volume but poor fulfilment outcomes?	Power BI	Guide seller coaching and operating reviews.
BQ-SAL-05	Sales	Which time periods show avoidable sales dips after campaign or stockout events?	Python	Plan campaign timing and inventory readiness.
BQ-SAL-06	Sales	Which cities should receive priority for profitable order expansion?	Machine Learning	Rank expansion opportunities by expected value.
BQ-SAL-07	Sales	How much of order growth comes from new customers versus repeat customers?	SQL	Balance acquisition and retention investment.
BQ-REV-01	Revenue	Which categories and sellers contribute the largest share of annual revenue?	SQL	Protect high-value revenue sources.
BQ-REV-02	Revenue	Which order value bands are growing fastest and how do they affect revenue mix?	Power BI	Track revenue quality by basket size.
BQ-REV-03	Revenue	Which discounts are increasing revenue but reducing net revenue quality?	Python	Tighten promotion guardrails.
BQ-REV-04	Revenue	How does revenue per active seller vary by category and region?	SQL	Improve seller productivity and marketplace mix.
BQ-REV-05	Revenue	Which customer acquisition channels generate the highest net revenue after returns?	Python	Shift budget toward cleaner revenue.
BQ-REV-06	Revenue	What revenue scenarios are required to reach the FY2026 growth target?	Python	Model growth path and executive targets.

Roadmap Notes

- Sales and revenue questions establish where growth is coming from and whether it is financially clean enough to support recovery targets.
- These questions should be answered before detailed margin bridges are finalised because they define the demand and revenue mix.

Profitability and Marketing Questions

These questions explain why growth is not translating into the required profit outcome.

ID	Domain	Business Question	Method	Roadmap Use
BQ-PRO-01	Profitability	Which categories create the largest gap between revenue growth and profit margin?	SQL	Find where growth is diluting margin.
BQ-PRO-02	Profitability	Which cost components explain the decline in profit despite revenue growth?	Python	Build the profit leakage bridge.
BQ-PRO-03	Profitability	Which orders have negative or low contribution after discounts, refunds and fulfilment cost?	SQL	Identify unprofitable order patterns.
BQ-PRO-04	Profitability	Which customer cohorts are profitable after acquisition, support and return costs?	Python	Refine retention and acquisition strategy.
BQ-PRO-05	Profitability	Which sellers should be prioritised for margin improvement actions?	Power BI	Support seller performance reviews.
BQ-PRO-06	Profitability	Can we predict which future orders are likely to be low contribution?	Machine Learning	Flag margin risk before fulfilment.
BQ-PRO-07	Profitability	Which levers have the highest expected impact on the 14% profit margin target?	Python	Rank recovery actions for leadership.
BQ-MKT-01	Marketing	Which campaigns deliver the strongest marketing ROI after returns and refunds?	SQL	Allocate campaign budget to profitable demand.
BQ-MKT-02	Marketing	Which paid channels have rising CAC without matching repeat purchase gains?	Power BI	Monitor inefficient acquisition channels.
BQ-MKT-03	Marketing	Which campaign cohorts show strong first orders but weak second-order conversion?	Python	Improve lifecycle targeting.
BQ-MKT-04	Marketing	Which promotions increase return rate or discount leakage?	SQL	Control promotion quality.
BQ-MKT-05	Marketing	Which customers are most likely to respond to retention offers?	Machine Learning	Prioritize offer targeting.
BQ-MKT-06	Marketing	Which marketing sources generate customers with the lowest support burden?	Python	Improve customer quality by source.

Roadmap Notes

- Profitability and marketing questions connect growth, campaign spend, acquisition efficiency, repeat purchase behaviour and margin leakage.
- The outputs should inform budget reallocation, promotion control and executive recovery priorities.

Customer and Product Questions

These questions connect customer behaviour, product quality, category mix and service experience.

ID	Domain	Business Question	Method	Roadmap Use
BQ-CUS-01	Customer Behaviour	Which customer segments have the highest repeat purchase rate?	SQL	Define retention priorities.
BQ-CUS-02	Customer Behaviour	Which customer journeys lead to abandoned carts or cancelled orders?	Python	Reduce shopping and checkout friction.
BQ-CUS-03	Customer Behaviour	Which delivery or return experiences reduce customer satisfaction most sharply?	Python	Target service recovery actions.
BQ-CUS-04	Customer Behaviour	Which customers are at risk of churn after delays, returns or support tickets?	Machine Learning	Create churn-risk intervention lists.
BQ-CUS-05	Customer Behaviour	How do new and repeat customers differ by category, payment type and city tier?	Power BI	Give leaders a customer-mix view.
BQ-CUS-06	Customer Behaviour	Which complaint themes are most strongly linked to lower repeat purchase?	Python	Prioritize customer experience fixes.
BQ-PRD-01	Product Performance	Which products generate high sales volume but high return rates?	SQL	Identify product quality and listing issues.
BQ-PRD-02	Product Performance	Which categories combine strong margin, low returns and strong repeat purchase?	Python	Prioritize category investment.
BQ-PRD-03	Product Performance	Which products create the highest support ticket volume?	SQL	Flag products needing content or quality review.
BQ-PRD-04	Product Performance	Which product attributes explain return or complaint patterns?	Machine Learning	Detect product risk drivers.
BQ-PRD-05	Product Performance	Which product categories should be featured in executive dashboard drilldowns?	Power BI	Improve dashboard navigation.
BQ-PRD-06	Product Performance	Which low-performing products should be delisted, improved or repriced?	Python	Support assortment decisions.

Roadmap Notes

- Customer and product questions explain whether service failures, product quality or assortment choices are affecting repeat purchase and satisfaction.
- These questions should feed customer-experience dashboards and product risk diagnostics.

Inventory and Warehouse Questions

These questions locate operational constraints before they become visible customer problems.

ID	Domain	Business Question	Method	Roadmap Use
BQ-INV-01	Inventory	Which SKUs and warehouses show the highest stockout rate?	SQL	Prioritize replenishment actions.
BQ-INV-02	Inventory	Which inventory ageing pockets are tying up working capital?	Power BI	Monitor ageing stock by category and node.
BQ-INV-03	Inventory	Which categories have weak forecast accuracy versus actual demand?	Python	Improve demand planning.
BQ-INV-04	Inventory	Can stockout risk be predicted before campaigns or seasonal peaks?	Machine Learning	Trigger proactive replenishment alerts.
BQ-INV-05	Inventory	Which warehouses should transfer inventory to reduce delays and lost sales?	Python	Support inventory transfer decisions.
BQ-INV-06	Inventory	How does inventory availability affect conversion and cancellation rates?	SQL	Link stock health to demand loss.
BQ-WHS-01	Warehouse	Which warehouses have the highest dispatch SLA breach rates?	SQL	Prioritize fulfilment process fixes.
BQ-WHS-02	Warehouse	Which order types create the most warehouse backlog?	SQL	Improve labour and slotting plans.
BQ-WHS-03	Warehouse	Which pick-pack stages cause the longest cycle time?	Python	Diagnose process bottlenecks.
BQ-WHS-04	Warehouse	Which warehouses contribute most to damaged or incorrect orders?	Power BI	Track fulfilment quality.
BQ-WHS-05	Warehouse	Can backlog risk be predicted from incoming orders, staffing and inventory status?	Machine Learning	Forecast fulfilment pressure.
BQ-WHS-06	Warehouse	Which warehouse improvements would most reduce average delivery time?	Python	Rank warehouse recovery actions.

Roadmap Notes

- Inventory and warehouse questions trace operational pressure before it reaches the customer through delays, cancellations or returns.
- The answers should guide replenishment, transfers, staffing, slotting and fulfilment process improvements.

Logistics and Returns Questions

These questions diagnose delivery reliability, reverse logistics and return-rate pressure.

ID	Domain	Business Question	Method	Roadmap Use
BQ-LOG-01	Logistics	Which courier partners have the highest SLA breach rate by region?	SQL	Support courier allocation decisions.
BQ-LOG-02	Logistics	Which routes or city tiers create the longest average delivery time?	Power BI	Monitor service-level hotspots.
BQ-LOG-03	Logistics	Which shipments are most likely to become return-to-origin cases?	Machine Learning	Prevent avoidable delivery failures.
BQ-LOG-04	Logistics	How much delivery delay is caused by warehouse dispatch versus courier transit?	Python	Separate internal and partner accountability.
BQ-LOG-05	Logistics	Which delivery delays lead to the largest CSAT decline?	Python	Prioritize customer-impacting SLA fixes.
BQ-LOG-06	Logistics	Which logistics exceptions should be shown in weekly operations dashboards?	Power BI	Standardize operating review views.
BQ-RET-01	Returns	Which categories and sellers drive the highest return rate?	SQL	Target seller and product quality actions.
BQ-RET-02	Returns	Which return reasons are increasing fastest after campaigns or delivery delays?	Python	Connect returns to operational triggers.
BQ-RET-03	Returns	Which return cases create the largest refund value leakage?	SQL	Reduce avoidable financial leakage.
BQ-RET-04	Returns	Can we predict orders likely to be returned before fulfilment?	Machine Learning	Flag high-risk orders and sellers.
BQ-RET-05	Returns	Which warehouses have the slowest return-to-stock cycle time?	Power BI	Monitor reverse logistics bottlenecks.
BQ-RET-06	Returns	Which return-policy exceptions affect repeat purchase behaviour?	Python	Balance customer recovery and control.

Roadmap Notes

- Logistics and returns questions separate partner performance, delivery reliability, reverse-logistics cost and customer-impact drivers.
- These questions are essential for reducing the 4.8-day delivery time and 12% return-rate problems established earlier.

Finance and Seller Questions

These questions connect financial controls, cost-to-serve, seller quality and marketplace governance.

ID	Domain	Business Question	Method	Roadmap Use
BQ-FIN-01	Finance	Which cost components are rising faster than revenue?	SQL	Identify financial pressure points.
BQ-FIN-02	Finance	Which settlements show recurring reconciliation variances?	SQL	Improve settlement controls.
BQ-FIN-03	Finance	How does cost-to-serve vary by category, warehouse, courier and customer segment?	Python	Build unit economics views.
BQ-FIN-04	Finance	Which refund and commission adjustments should be reviewed by finance?	Power BI	Create finance exception monitoring.
BQ-FIN-05	Finance	Which variables best explain refund leakage or settlement variance?	Machine Learning	Prioritize control-risk drivers.
BQ-FIN-06	Finance	What financial impact can be expected if return rate falls below 8%?	Python	Quantify target achievement value.
BQ-SEL-01	Seller Performance	Which sellers combine high revenue with poor customer experience outcomes?	SQL	Identify high-impact seller risks.
BQ-SEL-02	Seller Performance	Which sellers should receive coaching, restriction or renewal review?	Power BI	Support seller governance workflow.
BQ-SEL-03	Seller Performance	Which seller attributes are linked to high return or defect rates?	Machine Learning	Predict seller quality risk.
BQ-SEL-04	Seller Performance	How does seller dispatch performance affect delivery time and CSAT?	Python	Connect seller operations to customer trust.
BQ-SEL-05	Seller Performance	Which sellers underperform after controlling for category and region?	Python	Create fair seller benchmarking.
BQ-SEL-06	Seller Performance	Which seller KPIs should appear in the seller performance dashboard?	Power BI	Define dashboard scope and drilldowns.

Roadmap Notes

- Finance and seller questions reconcile cost, refund, commission and seller-quality issues with the profit recovery scorecard.
- The outputs should become exception views for finance control and seller governance.

Executive Reporting Questions

These questions translate functional analysis into executive scorecards and decision forums.

ID	Domain	Business Question	Method	Roadmap Use
BQ-EXE-01	Executive Reporting	Which KPIs require immediate executive attention this month?	Power BI	Prioritize leadership review.
BQ-EXE-02	Executive Reporting	What explains the gap between revenue growth and profit decline?	Python	Prepare executive profit bridge.
BQ-EXE-03	Executive Reporting	Which recovery initiatives are most likely to move FY2026 targets?	Machine Learning	Rank initiatives by expected impact.
BQ-EXE-04	Executive Reporting	What is the weekly status of revenue, margin, return rate, delivery time and CSAT?	Power BI	Standardize executive scorecard.
BQ-EXE-05	Executive Reporting	Which business owners are accountable for the largest KPI gaps?	SQL	Link metric gaps to action owners.
BQ-EXE-06	Executive Reporting	Which decisions should be escalated to the steering review?	Power BI	Convert analysis into executive actions.

Roadmap Notes

- Executive reporting questions turn detailed analysis into decision-ready scorecards, accountability views and steering-review actions.
- These questions should guide the first Power BI executive page and the recurring leadership readout.

Downstream Execution Focus

Future Workstream	Question Coverage	Expected Output
SQL data marts	KPI gap, owner accountability and source reconciliation questions.	Validated executive-summary marts.
Python diagnostics	Profit bridge, variance explanation and initiative scenario questions.	Reusable evidence notebooks and analysis narrative.
Power BI dashboards	Weekly scorecard, escalation and action-review questions.	Executive overview and action log pages.
Machine Learning	Recovery initiative ranking and early-warning prioritisation.	Candidate features for risk scoring and initiative ranking.